

Lab Activity 3.3: Construct form validation using Bootstrap and JavaScript

STUDENT INFORMATION

Name	Registration No	Date
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Learning Outcomes:

- a) Construct client-side form validation using JavaScript.
 - b) Construct Bootstrap form components for improved user experience.
 - c) Perform form submission and validation errors gracefully.
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Step 1: Setting Up the Document Structure & Styling

Create a new file named `passport_form.html`. We will start by defining the **Purple Theme** using CSS variables and styling the passport photo container to meet the **35mm x 45mm** requirement.

HTML

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Lab 3.3: Passport Application Form</title>
  <link
href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap
.min.css" rel="stylesheet">
<style>
  :root { --primary-purple: #6f42c1; --hover-purple: #59359a; --
light-purple: #f3e5f5; }
  body { background-color: var(--light-purple); padding: 40px 20px;
}
  .form-container { max-width: 800px; margin: auto; background:
white; padding: 40px; border-radius: 15px; border-top: 8px solid
var(--primary-purple); }
  .text-purple { color: var(--primary-purple); }
  .bg-purple { background-color: var(--primary-purple) !important;
}
  .btn-purple { background-color: var(--primary-purple); color:
white; }
```

```

        .btn-purple:hover { background-color: var(--hover-purple); color:
white; }
        .passport-photo { width: 3.5cm; height: 4.5cm; object-fit: cover;
border: 3px solid #f8f9fa; outline: 1px solid #dee2e6; display:
block; margin: 0 auto 10px auto; }
        .form-control:focus, .form-select:focus { border-color: var(--
primary-purple); box-shadow: 0 0 0 0.25rem rgba(111, 66, 193, 0.25);
}
        #rangeValue { background-color: var(--primary-purple); color:
white; padding: 2px 10px; border-radius: 10px; font-size: 0.9rem; }
        #successMessage { display: none; }
</style>
</head>

```

Step 2: Constructing the UI (Form Components)

Add the HTML body. This includes the **Success Alert**, **Passport Image**, **Floating Labels**, and **Input Groups**. Note the use of `novalidate` in the form tag to allow custom JavaScript handling.

HTML

```

<div class="container">
  <div class="form-container shadow-lg">

    <div class="mb-3">
      <a href="LAB2_1.html" class="text-decoration-none text-
purple small fw-bold">&larr; Back to Profile</a>
    </div>

    <h2 class="mb-4 text-center fw-bold text-purple">Passport
Application</h2>
    <hr class="mb-5">

    <div id="successMessage" class="alert alert-success alert-
dismissible fade show" role="alert">
      <strong>Success!</strong> Your application has been
validated and submitted gracefully.
      <button type="button" class="btn-close" data-bs-
dismiss="alert" aria-label="Close"></button>
    </div>

    <form class="row g-4 needs-validation" id="passportForm"
novalidate>

      <div class="col-12 text-center">
        
        <p class="text-muted small">Standard Passport Size
(35mm x 45mm)</p>
      </div>

      <div class="col-md-6">
        <div class="form-floating">

```

```

        <input type="text" class="form-control"
id="firstName" placeholder="First Name" required>
        <label for="firstName">First Name</label>
        <div class="invalid-feedback">First name is
required.</div>
    </div>
</div>

<div class="col-md-6">
    <div class="form-floating">
        <input type="text" class="form-control"
id="lastName" placeholder="Last Name" required>
        <label for="lastName">Last Name</label>
        <div class="invalid-feedback">Last name is
required.</div>
    </div>
</div>

<div class="col-md-12">
    <label for="userEmail" class="form-label fw-
bold">Contact Email</label>
    <div class="input-group has-validation">
        <span class="input-group-text bg-purple text-
white">@</span>
        <input type="email" class="form-control"
id="userEmail" placeholder="e.g. name@mail.com" required>
        <div class="invalid-feedback">Please enter a
valid email address.</div>
    </div>
</div>

<div class="col-md-6">
    <label for="citizenStatus" class="form-label fw-
bold">Citizenship</label>
    <select class="form-select" id="citizenStatus"
required>
        <option selected disabled value="">Select
status...</option>
        <option>Citizen</option>
        <option>Permanent Resident</option>
    </select>
    <div class="invalid-feedback">Selection
required.</div>
</div>

<div class="col-md-6">
    <label for="yearsResidency" class="form-label fw-
bold">
        Residency Years: <span id="rangeValue">25</span>
    </label>
    <input type="range" class="form-range" min="1"
max="50" value="25" id="yearsResidency">
</div>

<div class="col-md-12">
    <label class="form-label d-block fw-
bold">Gender</label>
    <div class="form-check form-check-inline">

```

```

        <input class="form-check-input" type="radio"
name="gender" id="male" required>
        <label class="form-check-label"
for="male">Male</label>
    </div>
    <div class="form-check form-check-inline">
        <input class="form-check-input" type="radio"
name="gender" id="female" required>
        <label class="form-check-label"
for="female">Female</label>
    </div>
    <div class="invalid-feedback">Please select a
gender.</div>
</div>

    <div class="col-12">
        <div class="form-check">
            <input class="form-check-input" type="checkbox"
value="" id="invalidCheck" required>
            <label class="form-check-label" for="invalidCheck">
                I confirm that all information provided is
accurate.
            </label>
            <div class="invalid-feedback">You must agree before
submitting.</div>
        </div>
    </div>

    <div class="col-md-6">
        <div class="form-floating">
            <input type="password" class="form-control"
id="password" placeholder="Password" required>
            <label for="password">Password</label>
            <div class="invalid-feedback">Please enter a
password.</div>
        </div>
    </div>

    <div class="col-md-6">
        <div class="form-floating">
            <input type="password" class="form-control"
id="confirmPassword" placeholder="Confirm Password" required>
            <label for="confirmPassword">Confirm
Password</label>
            <div class="invalid-feedback"
id="confirmFeedback">Passwords do not match!</div>
        </div>
    </div>

    <div class="col-12 mt-5 d-flex gap-2">
        <button class="btn btn-purple w-75 py-3 fw-bold
shadow-sm" type="submit">SUBMIT APPLICATION</button>
        <button class="btn btn-outline-secondary w-25 py-3
fw-bold shadow-sm" type="reset" id="resetBtn">RESET</button>
    </div>

</form>
</div>

```

```
</div>
```

Step 3: JavaScript Logic (Validation & Interaction)

Implement the script to handle **Range Slider** updates, **Password Matching**, and **Graceful Error Handling**.

HTML

```
<script>
  (function () {
    'use strict'

    const form = document.getElementById('passportForm');
    const successDiv = document.getElementById('successMessage');
    const resetBtn = document.getElementById('resetBtn');
    const rangeInput = document.getElementById('yearsResidency');
    const rangeValue = document.getElementById('rangeValue');

    // Update range value display
    rangeInput.addEventListener('input', function() {
      rangeValue.textContent = this.value;
    });

    // Form Validation Logic
    form.addEventListener('submit', function (event) {
      if (!form.checkValidity()) {
        event.preventDefault();
        event.stopPropagation();
        successDiv.style.display = 'none';
      } else {
        event.preventDefault();
        successDiv.style.display = 'block';
        window.scrollTo({ top: 0, behavior: 'smooth' });
      }
      form.classList.add('was-validated');
    }, false);

    // Reset Logic
    resetBtn.addEventListener('click', function() {
      form.classList.remove('was-validated');
      successDiv.style.display = 'none';
      rangeValue.textContent = "25"; // Reset range display to
default
    });

  })();
  (function () {
    'use strict'

    const form = document.getElementById('passportForm');
    const password = document.getElementById('password');
    const confirmPassword =
document.getElementById('confirmPassword');

    // Fungsi untuk menyemak padanan kata laluan
```

```

function validatePassword() {
    if (password.value !== confirmPassword.value) {
        // Jika tidak sama, set ralat custom
        confirmPassword.setCustomValidity("Passwords do not
match");
    } else {
        // Jika sama, kosongkan ralat (sah)
        confirmPassword.setCustomValidity("");
    }
}

// Jalankan fungsi setiap kali pengguna menaip
password.addEventListener('change', validatePassword);
confirmPassword.addEventListener('keyup', validatePassword);

// Kekalkan logik submit anda yang sedia ada
form.addEventListener('submit', function (event) {
    // validatePassword dipanggil sekali lagi sebelum submit
untuk kepastian
    validatePassword();

    if (!form.checkValidity()) {
        event.preventDefault();
        event.stopPropagation();
    } else {
        event.preventDefault();
        document.getElementById('successMessage').style.display =
'block';
        window.scrollTo({ top: 0, behavior: 'smooth' });
    }
    form.classList.add('was-validated');
}, false);

// ... (Logik Range dan Reset anda yang lain)
})();
</script>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.b
undle.min.js"></script>

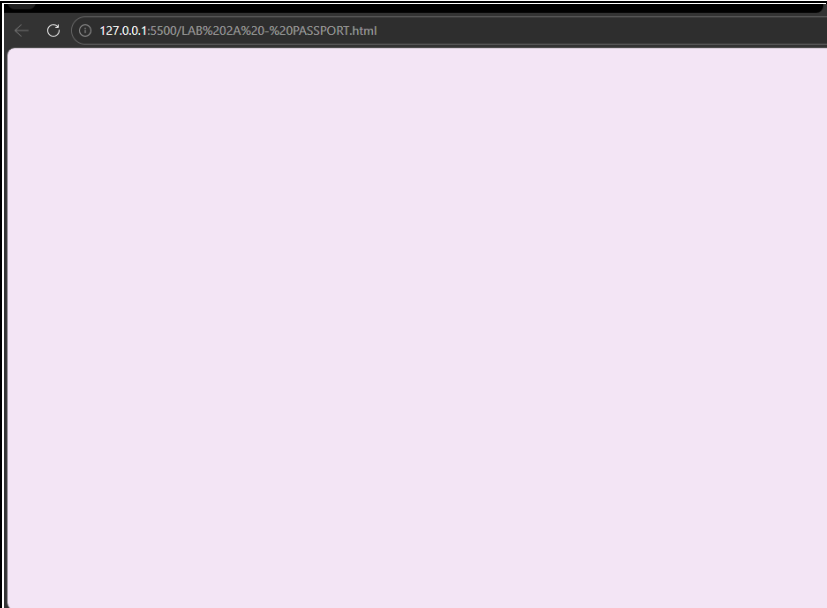
```

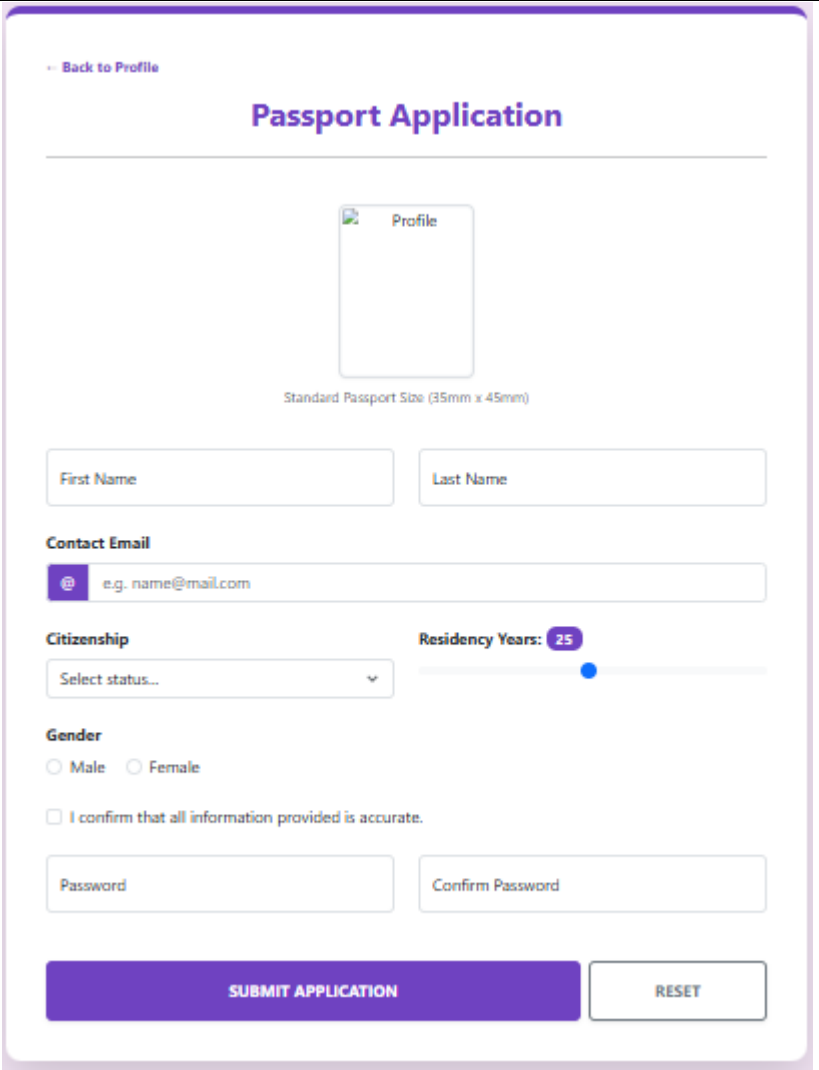
Task Recap for Students:

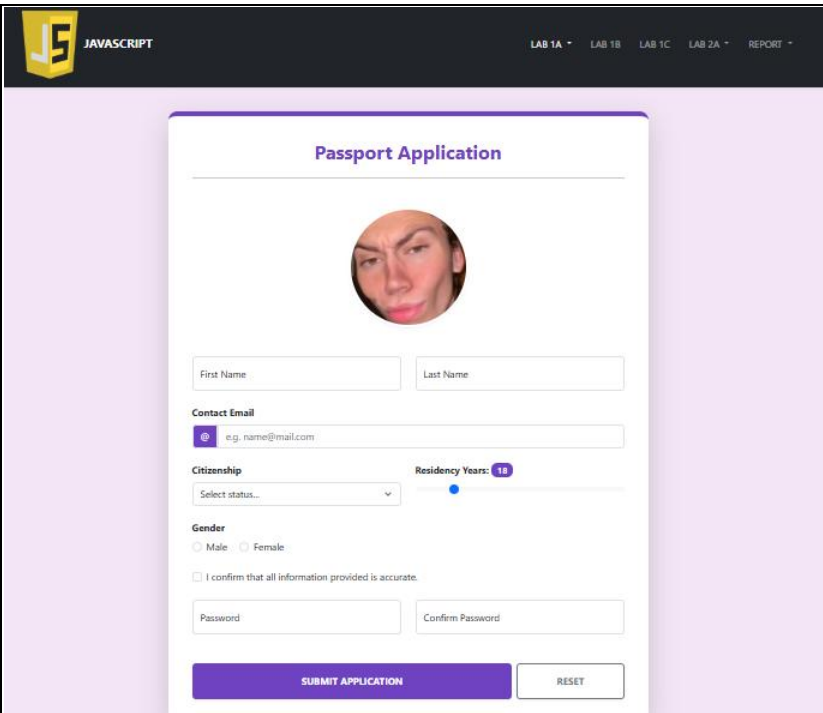
1. **Step 1:** Copy CSS to define the look and the specific **3.5cm x 4.5cm** image size.
2. **Step 2:** Construct the form using Bootstrap's grid and specialized components.
3. **Step 3:** Apply the JavaScript to enable live slider updates and ensure the passwords match before submission.
4. **Observation:** Notice how the `was-validated` class triggers red/green borders and messages gracefully without refreshing the page.

STEP-BY-STEP VALIDATION:

Please provide a **screenshot** for every completed step. For each image, write a short **observation** explaining the logic behind the code you just implemented. This helps solidify your mastery of Bootstrap!

Progress Step	Screenshot of Output	Your Observation & Understanding
Step 1	 A screenshot of a web browser window. The address bar shows the URL "127.0.0.1:5500/LAB%202A%20-%20PASSPORT.html". The main content area of the browser is a solid light purple color.	Penetapan tema dilakukan menggunakan CSS <i>variables</i> (:root) untuk memastikan konsistensi warna ungu di seluruh borang. Kontena gambar pasport juga ditetapkan pada saiz spesifik 35mm x 45mm mengikut piawaian rasmi.

Progress Step	Screenshot of Output	Your Observation & Understanding
Step 1		Penggunaan komponen Bootstrap seperti <i>Floating Labels</i> dan <i>Input Groups</i> meningkatkan estetika dan kemudahan pengguna (UX). Atribut <code>novalidate</code> diletakkan pada tag <code><form></code> untuk membolehkan JavaScript mengendalikan ralat pengesahan secara tersuai.

Progress Step	Screenshot of Output	Your Observation & Understanding
<i>Cont...</i>	 The screenshot shows a web browser window with a dark header. On the left is a yellow JavaScript logo. On the right are navigation links: LAB 1A, LAB 1B, LAB 1C, LAB 2A, and a REPORT link. The main content area has a light purple background. Centered on the page is a white 'Passport Application' form. The form includes a circular profile picture of a man. Below the picture are input fields for 'First Name' and 'Last Name'. A 'Contact Email' field contains the placeholder 'e.g. name@mail.com'. There is a 'Citizenship' dropdown menu set to 'Select status...' and a 'Residency Years' slider set to 18. The 'Gender' section has radio buttons for 'Male' and 'Female'. A checkbox for 'I confirm that all information provided is accurate.' is present. At the bottom are 'Password' and 'Confirm Password' fields. Two buttons, 'SUBMIT APPLICATION' and 'RESET', are at the very bottom of the form.	

5. SCREENSHOT OUTPUT (COMPLETE):

Desktop View

Passport Application

First Name Last Name

Contact Email
e.g. name@mail.com

Citizenship
Select status...

Residency Years: 10

Gender
☐ Male ☐ Female

☐ I confirm that all information provided is accurate.

Password Confirm Password

SUBMIT APPLICATION RESET

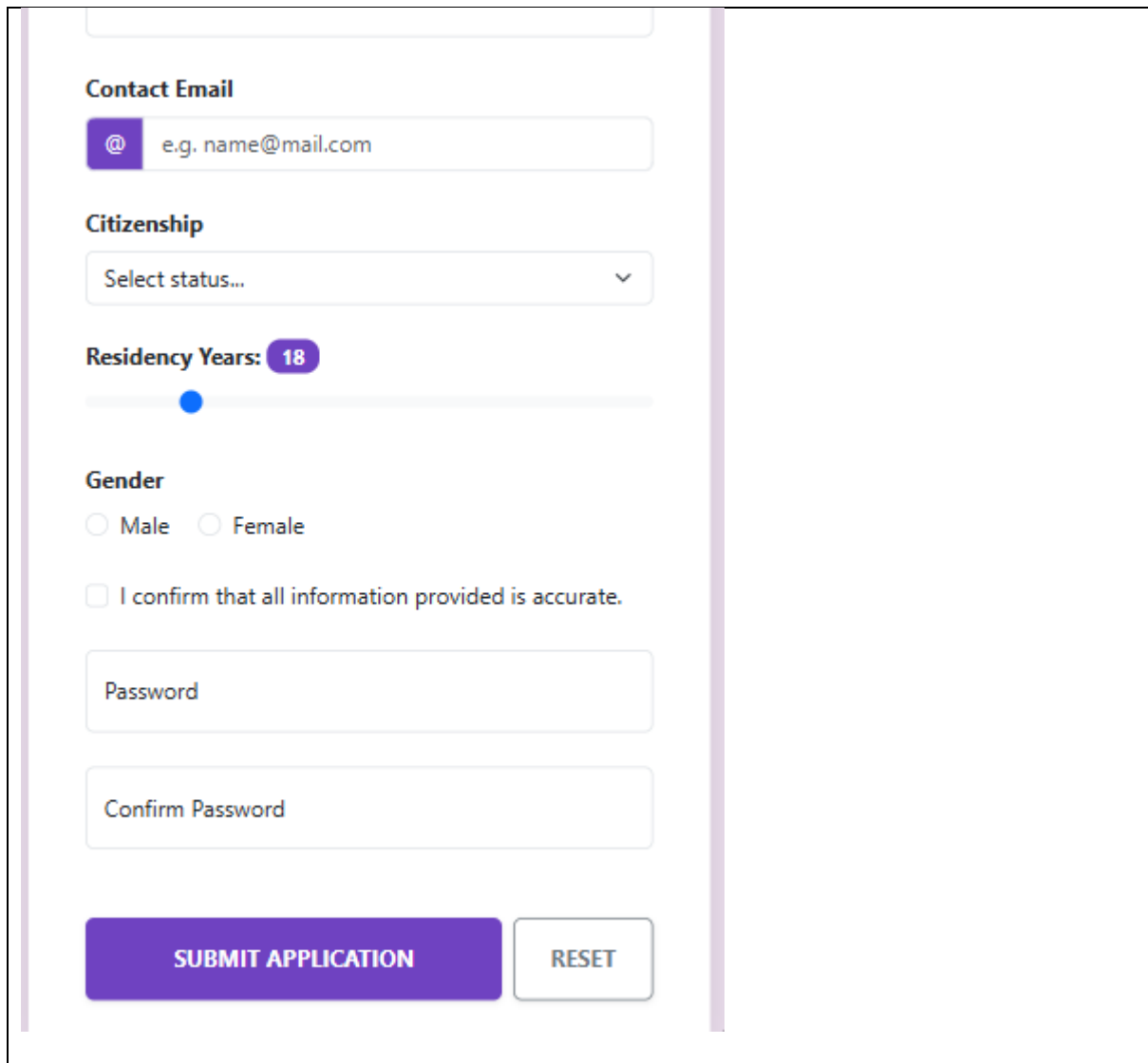
Mobile View

JAVASCRIPT



Passport Application



A registration form interface with a light purple border. It contains several input fields and a slider. The 'Contact Email' field has a purple icon with an '@' symbol and placeholder text 'e.g. name@mail.com'. The 'Citizenship' field is a dropdown menu with 'Select status...' and a downward arrow. The 'Residency Years' field shows a value of '18' in a purple circle next to a horizontal slider bar with a blue dot. The 'Gender' section has two radio buttons for 'Male' and 'Female'. Below them is a checkbox for 'I confirm that all information provided is accurate.'. There are two password fields labeled 'Password' and 'Confirm Password'. At the bottom are two buttons: a purple 'SUBMIT APPLICATION' button and a white 'RESET' button with a purple border.

Contact Email

@ e.g. name@mail.com

Citizenship

Select status... ▼

Residency Years: 18

Gender

☐ Male ☐ Female

☐ I confirm that all information provided is accurate.

Password

Confirm Password

SUBMIT APPLICATION RESET

6. LAB SUBMISSION TASK

Ensure you have completed all steps successfully!

1. **Compress** your folder: RegNo_Date.zip.
2. **Report:** Create a PDF containing all screenshots and explanations:
Report_RegNo_Date.pdf.

Happy Coding! Success is in your hands!